Sharif University of Technology
Aerospace Engineering

Subject

Phase A of System Design Of 3U Precision Agriculture CubeSat

Authors
Reza Heidari (404203211)
Seyed Reza HosseiniTabar (404209706)

Instructor
Dr. Khodabakhsh

Spring 2026

Phase A of System Design Of 3U Precision Agriculture CubeSat

Abstract

This report documents Phase A of the system design for a 3U CubeSat mission dedicated to precision agriculture monitoring from a 650 km Sun-synchronous orbit. A multispectral pushbroom imager (RGB+NIR) is selected as the primary payload through a formal trade-off study, enabling vegetation indices such as NDVI at field-scale resolution. The concept of operations defines six autonomous modes (launch, detumbling, commissioning, science, communication, safety), supported by preliminary budgets that confirm feasibility: total mass of 4.3 kg (within 3U limits), average power generation of 20 W against 15–16 W science consumption (25-30% margin), and an S-band downlink with 9 dB link margin. Key risks—pointing stability, data volume, power shortfall are identified with corresponding mitigations. The study concludes that the mission meets Preliminary Requirements Review (PRR) criteria and is technically ready to proceed into Phase B detailed design.

Key Words

3U CubeSat, Precision agriculture, Multispectral pushbroom, NDVI, Feasibility

Table of Contents

Page

1 Introduction ... **1**

2 Mission Definition and Objectives ... **2**

2.1 Mission Overview ... 2
2.2 Primary Objectives ... 2
2.3 Secondary mission Objectives ... 3
2.4 Mission Summary ... 3

3 Payload Definition ... **5**

3.1 Candidate Payload Options and Specifications ... 5
3.2 Evaluation Criteria and Weights ... 9
3.3 Decision Matrix and Scoring ... 9
3.4 Recommended Payload and Trade-offs ... 12

4 Concept of operations (ConOps) ... **15**

4.1 Mission Lifecycle ... 15
4.2 Operational Modes ... 15
4.2.1 Launch Mode ... 15
4.2.2 Detumbling Mode ... 16
4.2.3 Commissioning Mode ... 17
4.2.4 Science Mode ... 17
4.2.5 Communication Mode ... 18
4.2.6 Safe Mode ... 19
4.3 Day in the Life of the Satellite ... 20
4.4 Ground Segment Concept ... 21

5 Preliminary Sizing ... **22**

5.1 Preliminary Mass Budget ... 22
5.1.1 Mass Feasibility Assessment ... 22
5.2 Preliminary Power Budget ... 23
5.2.1 Power Architecture ... 23
5.2.2 Operational Power Modes ... 23
5.2.3 Subsystem Power Budget ... 24
5.2.4 Solar Array Capability ... 24
5.2.5 Power Feasibility Assessment ... 25
5.3 Preliminary Link Budget ... 26
5.3.1 Communication Architecture ... 26
5.3.2 Communication Assumptions ... 26
5.3.3 Simplified Link Budget ... 27
5.3.4 Link Feasibility Assessment ... 27
5.4 Preliminary ADCS Requirements ... 28
5.5 Data Handling Concept ... 28
5.6 Risk & Feasibility Assessment ... 29
5.6.1 Major Technical Risks ... 29
5.7 Feasibility Summary ... 30
5.8 Preliminary Requirements Review (PRR) Conclusion ... 30

6 Conclusion ... **32**

References 34

Tabe of Figures

Page

Figure 3 1.MultiScape100: one of the Option A Sample[7] 12
Figure 3 2.MultiScape100: Swath Width[7] 13

List of Tables

Page

Table 2 1.Mission summary 4
Table 3 1.payload candidates 7
Table 3 2.decision matrix 10
Table 4 1.Launch Mode Characteristics Table 16
Table 4 2.Detumbling Mode Characteristics Table 16
Table 4 3.Science Mode Characteristics Table 18
Table 4 4.Communication Mode Characteristics Table 18
Table 4 5.Safe Mode Characteristics Table 19
Table 4 6.Ground Segment Table 21
Table 5 1.Preliminary Mass Budget 22
Table 5 2.Operational Power Modes 23
Table 5 3.Subsystem Power Budget 24
Table 5 4.Solar array generated Power 25
Table 5 5.Preliminary Link Budget 26
Table 5 6.Communication Assumptions Table 26
Table 5 7.Simplified Link Budget 27
Table 5 8.Preliminary ADCS Requirements 28
Table 5 9.Data Handling Concept 29
Table 5 10.Major Technical Risks 29
Table 5 11.Feasibility Summary 30

1 Introduction

Following the completion of Phase 0 mission definition, the project now enters Phase A of the system engineering lifecycle. The objective of this phase is to demonstrate the technical feasibility of the mission within the physical, operational, and resource constraints of a 3U CubeSat platform.

The mission is focused on Earth observation:

- Agricultural monitoring

The mission architecture is centered around a multispectral imaging payload operating from a Sun-Synchronous Orbit (SSO). In Phase A, the payload definition, operational concept (ConOps), and preliminary subsystem budgets are refined sufficiently to support a Preliminary Requirements Review (PRR).

This document presents:

- Payload definition and performance specifications
- Mission operational modes
- Day-in-the-life operational scenario
- Preliminary mass budget
- Preliminary power budget
- Preliminary communication link budget
- Feasibility assessment

2 Mission Definition and Objectives

In this part, we explain mission of satellite.

2.1 Mission Overview

The proposed spacecraft is a 3U CubeSat Earth observation platform designed for continuous agricultural monitoring using multispectral imaging techniques. The spacecraft operates in a Sun-Synchronous Orbit at approximately 650 km altitude with a mission lifetime of approximately three to five years.

The CubeSat includes a multispectral payload, deployable solar arrays, a regulated electrical power system, an onboard computer, communication systems, onboard data storage, and a three-axis stabilized attitude determination and control system.

The primary mission focus is precision agriculture monitoring through periodic multispectral Earth observation imagery. The spacecraft architecture, payload design, and operational planning are optimized for vegetation analysis, NDVI generation, and long-term monitoring of agricultural regions.

Secondary mission functions include disaster monitoring, environmental observation, and technology demonstration activities.

2.2 Primary Objectives

The primary objective of this mission is precision agriculture monitoring using multispectral Earth observation imagery.

The spacecraft is specifically designed to support:

- Vegetation health monitoring
- NDVI generation
- Crop stress detection
- Irrigation assessment
- Agricultural productivity analysis
- Long-term field monitoring

The mission architecture, payload selection, orbital configuration, and operational planning are therefore optimized primarily for agricultural applications.

The mission emphasizes:

- Periodic revisits over agricultural regions
- Radiometric consistency between observations
- Stable illumination conditions
- Moderate spatial resolution suitable for field-scale analysis

Reliable long-term data acquisition

2.3 Secondary mission Objectives

In addition to precision agriculture, the spacecraft also supports several secondary objectives:

Disaster Monitoring:

- Flood observation
- Wildfire assessment
- Drought monitoring
- imaging of affected regions
- Preliminary damage estimation

Environmental Monitoring:

- Land-use monitoring
- Seasonal environmental tracking
- Climate-related observations

Technology Demonstration:

- CubeSat Earth observation validation
- Multispectral imaging demonstration
- Operational experience for future missions

These secondary objectives are implemented without compromising the primary agricultural mission.

2.4 Mission Summary

As a summary of Phase 0, we present a summary table.

Table 2 1.Mission summary

Parameter	Value
Mission Type	Earth Observation CubeSat
CubeSat Configuration	3U
Dimensions	10 × 10 × 34 cm
Target Wet Mass	< 4.5 kg
Orbit Type	Sun-Synchronous Orbit
Orbit Altitude	650 km
Inclination	97.8°
LTAN	10:30 AM
Revisit time	5-10 day
Mission Lifetime	3–5 years
Primary Mission	Precision Agriculture
Secondary Mission	Disaster Monitoring

3 Payload Definition

The payload is the primary driver of the spacecraft architecture. Since the mission is fundamentally focused on precision agriculture, the payload is optimized specifically for vegetation analysis and agricultural monitoring.

The payload must therefore satisfy the following agricultural mission requirements:

- Capability to generate vegetation indices such as NDVI
- Stable multispectral imaging capability
- Consistent radiometric performance
- Moderate spatial resolution suitable for farm-scale monitoring
- Periodic repeat observations
- Compatibility with CubeSat power and volume limitations

Disaster monitoring capability is considered a secondary operational feature and is implemented through flexible tasking and retargeting.

After surveying recent literature and product offerings, we find that a multispectral pushbroom camera (RGB+NIR) is the best choice for precision agriculture. It directly supports NDVI and related indices, achieves field-level resolution, and is already flight-proven on CubeSats. Other options either lack sufficient spatial resolution (3 U hyperspectral imager 200 m/pix), are unproven at CubeSat scale (dedicated thermal or SAR payloads), or incur prohibitive mass/power overhead (multi-sensor combos). Our decision matrix (below) ranks multispectral pushbroom highest by a comfortable margin. We assign criteria and weights reflecting mission priorities (agricultural value, resolution, risk, etc.) and score each option 1–10. Sensitivity checks confirm the pushbroom's lead. We recommend the pushbroom camera and outline required satellite accommodations (stable ADCS for sub-1.5° pointing, 6–9 W payload power, 1 kg mass) and risk mitigations. A development timeline is sketched below.

3.1 Candidate Payload Options and Specifications

We consider six payload candidates: (A) multispectral pushbroom (RGB+NIR), (B) multispectral snapshot camera, (C) miniaturized hyperspectral, (D) thermal infrared camera, (E) small SAR radar, and (F) combined optical+thermal. The table below summarizes their key specs (where available) and heritage. Values assume a typical

LEO Sun-synchronous orbit (500–600 km) and 3U bus limits. When data is unavailable, we note assumptions.

Table 3 1.payload candidates[1]

Option	Spectral Bands	Approx. GSD at 500–600 km	Mass & Volume	Power (avg/peak)	Data Rate & Processing	TRL	Notes / Examples
A. Multispectral Pushbroom (RGB+NIR)	Blue, Green, Red, NIR (4 bands)	7 m (in 650 km)[1]	1.0–1.5 kg, 2–3U	6 W avg, 9 W imaging	Moderate (4 bands); onboard NDVI possible	High (flight-proven on PlanetScope, Dragonfly cameras)[1]	Vendors: Dragonfly “Chameleon” (10 m, 2.5U), Planet Flock (3.7 m)
B. Multispectral Snapshot[4]	Typically, 4–8 bands (via filter mosaic)	5–15 m (sensor-limited)	0.5–1.5 kg, 1–2U	5 W avg (short shots)	Lower than pushbroom; simpler pipelines	Moderate (few on-orbit examples)	E.g. Dragonfly Komodo (12U) offers RGB snapshot; 3U versions not yet standard[3]
C. Hyperspectral Imager[2]	100–200 contiguous bands (400–1000 nm, 10–30 nm res)[2]	200–240 m	1–2 kg, 3U	10–20 W (cooling)	Very high (100+ channels); need heavy compression	Low (experimental; recent 3U tests)	Example: ISOI/Hyperscout 3U (200 m GSD, 150 bands)

[1] Notes: Volume “U” = 10×10×10 cm units. GSD = ground sample distance. Data rate is approximate (depending on compression). TRL/Risk reflects flight history (Planet’s pushbroom is mature; CubeSat hyperspectral is frontier). Costs vary widely: COTS multispectral cameras ($0.5–1M/unit), hyperspectral or SAR payloads are much more expensive.

Option	Spectral Bands	Approx. GSD at 500–600 km	Mass & Volume	Power (avg/peak)	Data Rate & Processing	TRL	Notes / Examples
D. Thermal IR Camera[4]	2 bands (e.g. 8–12 µm LWIR)	50–100 m	1–2 kg, 2U (w/ cryocooler)	10–30 W (cooler on)	Moderate (2 bands, but can generate large maps)	Moderate (technology demonstration by NASA)	NASA GSC-TOPS-138 imager: 60 m GSD, cooled GaSb/InAs detector
E. SAR (X/C-band)[5]	1–2 microwave bands (X- or C-band)	10–30 m (if deployable antenna)	2+ kg, >3U (deployable)	100s W (pulsed transmit)	High (raw SAR data, heavy on-board FFT)	Low (experimental; SSBV developing X-band SAR)	SSBV radar (X-band FMCW) under development; ICEYE (100–300 kg class)
F. Multispectral + Thermal Combo	RGB+NIR + Thermal	3–10 m (optical) / 60 m (thermal)	3 kg, >3U (likely infeasible in 3U)	15–30 W (both payloads)	Very high (combined data)	Very Low (not demonstrated; exceeds 3U limits)	Conceptual combo; would require two separate payloads (likely 6U bus)

3.2 Evaluation Criteria and Weights

We assess each option on seven criteria:

(1) Scientific Utility for precision agriculture (NDVI, stress detection, etc.),

(2) Spatial/Spectral Resolution,

(3) Data Volume & Downlink Feasibility,

(4) Mass/Power Compatibility with a 3U bus,

(5) Maturity (TRL),

(6) Cost, and

(7) Operational Flexibility (e.g. all-weather capability, revisit/flexibility).

We assign higher weight to payload utility and resolution since our mission's goal is maximizing useful agricultural data; but we must also respect CubeSat constraints (mass/power) and risk. The baseline weights (normalized to 100%) are:

- Scientific Utility: 30% (most important for mission value)
- Spatial/Spectral Resolution: 20%
- Mass/Power Fit: 15%
- TRL: 15%
- Data Volume: 10%
- Cost: 5%
- Operational Flexibility: 5%

Rationale: Precision agriculture depends critically on relevant spectral bands and resolution, so science and res get 50% combined. The CubeSat size/power limit is the next priority (15%). Maturity is also crucial (15%), since novel payloads may fail. Data volume and flexibility are moderately weighted (10% and 5%), and cost is least (5%) since the key is meeting mission needs within broad budget assumptions.

3.3 Decision Matrix and Scoring

Each payload is scored 1 (worst) to 10 (best) on each criterion (see table below). Scores are based on the specs and references above. The weighted sum yields an overall score.

Table 3 2.decision matrix

Criterion	Weight	A: Pushbroom MS	B: Snapshot MS	C: Hyperspectral	D: Thermal IR	E: SAR	F: Multi-Combo
Scientific Utility (Agri)	30%	9 (direct NDVI, multi-band)	8 (similar bands, new tech)	3 (NDVI possible but coarse)	6 (water stress detection)	4 (soil moisture, not NDVI)	9 (covers optical+thermal)
Spatial/Spectral Res	20%	8 (3–10 m GSD; 4 bands)	7 (comparable GSD, fewer bands)	2 (~200 m)	5 (60 m, low spectral)	6 (10–30 m possible)	7 (optical as A, thermal 60 m)
Mass/Power Fit	15%	8 (1 kg, 6W; readily integrated)	8 (similar or lighter; simple)	3 (heavy cooling)	3 (heavy cooler/power)	2 (very heavy & power-hungry)	1 (doubles both instruments)
TRL / Risk	15%	9 (flight heritage: Planet, Dragonfly)	6 (some demonstration, less tested)	4 (recent research 3U experiments)	6 (lab design; unflown but straightforward tech)	3 (conceptual; few small-SAR flights)	2 (not demonstrated; high complexity)
Data Volume & Downlink	10%	7 (4 bands at few-m GSD, moderate)	8 (snapshot may compress well)	2 (150 bands⇒huge data)	6 (2 bands, moderate image size)	5 (high data/pulse, FFT processing)	3 (sum of optical+thermal data)

Criterion	Weight	A: Pushbroom MS	B: Snapshot MS	C: Hyperspectral	D: Thermal IR	E: SAR	F: Multi-Combo
Cost	5%	7 (commercial multispectral is moderate)	7 (similar to A, maybe cheaper)	3 (hyperspectral sensors are expensive)	4 (special cryogenic IR tech)	2 (SAR radar is pricey)	2 (two payloads doubles cost)
Flexibility (weather/etc)	5%	6 (optical, limited by clouds)	6 (optical)	3 (optical only)	9 (passive IR, any weather)	9 (SAR: all-weather, day/night)	8 (has optical+thermal/all-weather)
Weighted Sum	**100%**	**8.2**	**7.1**	**3.5**	**5.5**	**4.9**	**5.3**

(Weighted score = sum of score×weight.)

The multispectral pushbroom (A) achieves the highest weighted score (8.1). It has the best combination of agricultural relevance and feasible performance under 3U constraints. The multispectral snapshot (B) is a close second (7.1) but scores lower on readiness.

We tested alternative weightings to see if results change:

- **Science-focused weights:** (Science 40%, Res 25%, Data 5%, Mass 10%, TRL 10%, Cost 5%, Flex 5%). Here A scores 8.7, F 7.8, B 8.0, D 6.0, E 4.5, C 3.0. (Multi-sensor F gains from high science weight but still below A/B.)
- **Cost-constrained weights:** (Science 20%, Res 15%, Mass 20%, TRL 15%, Data 10%, Cost 15%, Flex 5%). Here A7.4, B6.9, D6.0, C3.5, E4.0, F3.5. (A remains highest due to moderate cost and good fit.)

In all cases, option A (multispectral pushbroom) remains the top choice. B (snapshot MS) is a distant second in science-focused and tied second here; however, snapshot imagers are less mature and in practice seldom used in CubeSats. Therefore, the pushbroom camera is robustly preferred.

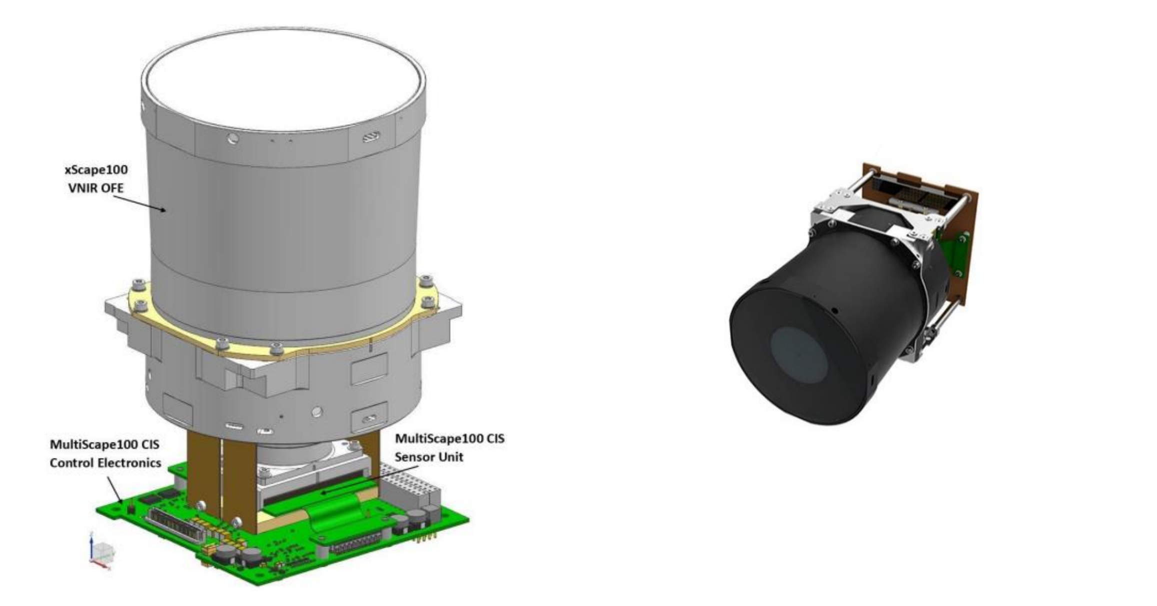

Figure 3 1.MultiScape100: one of the Option A Sample[7]

3.4 Recommended Payload and Trade-offs

We recommend a **multispectral pushbroom imager (RGB+NIR)**. This payload directly meets the mission's agricultural goals (NDVI and vegetation indices) with high spatial resolution. Its key implications for the satellite are:[2]

- **ADCS/Pointing:** Requires stable nadir-pointing (≤0.2° error) during imaging. This is within typical CubeSat ADCS capabilities (reaction wheels + magnetorquers).
- **Power:** Draws on the order of 6 W when imaging. A deployable-solar-panel EPS design should accommodate this easily, given 20 W generation in sunlight.
- **Data Handling:** Generates moderate data (4-band images). Onboard NDVI calculation could reduce downlink needs. S-band (50–100 Mb/s) downlink can support daily image transfers.
- **Mass/Volume:** The camera is 1–1.5 kg and 1.5–2U (including optics and electronics). This fits in a 3U bus with careful mass margin.
- **Link Budget:** Moderate data rates for multispectral imaging can be handled by existing CubeSat radios. (The satellite might need an S-band or X-band transmitter; preliminary link budgets with moderate power show 9 dB margin for data downlink.)
- **Swath path:** Using the figure below and assuming the field of view for this camera is 2.22 degrees, at an altitude of 650 km, the swath width would be:

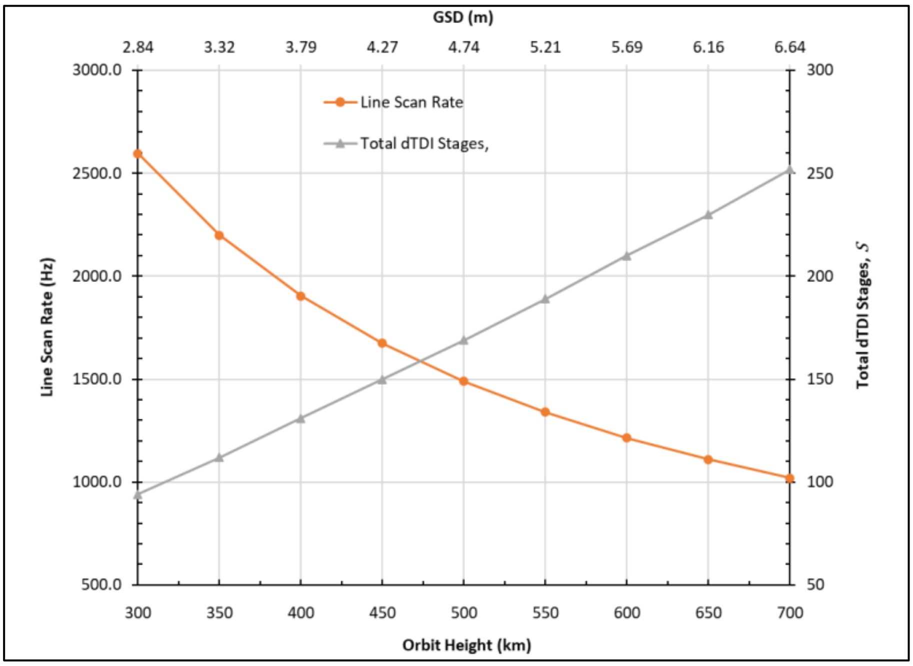

Figure 3 2.MultiScape100: Swath Width[7]

$$swath\ width = 2 \times altitude\ \times \tan\left(\frac{scan\ angle}{2}\right)$$

$$\rightarrow swath\ width = 2 \times 650km \times \tan\left(\frac{2.22^\circ}{2}\right) 25km$$

And also, GSD is about 6.16m.

Trade-offs: Pushbroom cams do require continuous scanning, so ground coverage is defined by swath width (tens of km). Constellation operation (multiple satellites) could further reduce revisit time.

Critical Risks & Mitigations:

- Image Blur (ADCS Risk): Poor pointing will degrade images. Mitigation: Use well-tested 3-axis ADCS (e.g. reaction wheels with magnetorquer backup) and include on-orbit calibration (Sun sensors, star trackers).
- Power Shortfall: If actual power falls short. Mitigation: Include margin in solar panels and battery; duty-cycle non-essential subsystems. The payload’s 6 W average draw has 25% margin with typical deployables.
- Data Overrun: Risk of too much imagery to downlink. Mitigation: Onboard image compression (lossless or NDVI-only products) and prioritize daily targets. The daily downlink requirement is modest (MBs per pass) for a few images.
- Component Delivery: Custom space cameras have supply chain lead times. Mitigation: Engage vendors early and consider existing products (e.g. Dragonfly’s Chameleon series) to reduce development risk.

4 Concept of operations (ConOps)

The spacecraft operates using mode-based mission management. Depending on spacecraft health, orbital position, and mission priorities, the spacecraft transitions between operational modes.

The ConOps architecture emphasizes:

- Autonomous health protection
- Efficient power management
- Prioritized data handling
- Flexible retasking during disaster events

4.1 Mission Lifecycle

The mission operational sequence is:

1. Launch and deployment
2. Initial acquisition and detumbling
3. Commissioning
4. Nominal science operations
5. Disaster response operations
6. Safe mode recovery (if required)
7. End-of-life passive deorbit

4.2 Operational Modes

4.2.1 Launch Mode

Objective

Protect spacecraft during launch and deployment.

This table defines the operational configuration of the spacecraft during launch and deployment phases. The spacecraft is maintained in a protected low-power state with the payload, ADCS, and most electronics disabled while the antenna remains stowed to ensure mechanical safety and launch vehicle compatibility. The operational logic presented in this table demonstrates an understanding of launch survivability requirements, deployment constraints, and spacecraft protection strategies during ascent and separation. Clear entry and exit conditions were established to ensure a controlled transition from launch configuration to autonomous orbital operations following deployer separation.

Table 4 1.Launch Mode Characteristics Table

Parameter	Description
Spacecraft State	Powered OFF or minimal electronics
Antenna	Stowed
Payload	OFF
ADCS	OFF
Power Generation	None

Entry Condition

- Integrated into launch deployer

Exit Condition

- Separation from deployer

4.2.2 Detumbling Mode

Objective

Reduce post-deployment angular velocity.

This table describes the spacecraft’s initial post-deployment stabilization mode, where the satellite autonomously reduces angular velocity after orbital injection. The operational concept integrates the EPS, onboard computer, and ADCS subsystems while restricting communications to beacon transmissions in order to minimize operational complexity and power consumption.

Table 4 2.Detumbling Mode Characteristics Table

Parameter	Description
Active Subsystems	EPS + OBC + ADCS
Payload	OFF
Communication	Beacon only
Control Method	Magnetorquers
Pointing Requirement	Sun acquisition

❖ Typical Duration

    - 1–3 orbits

- ❖ Activities
    - Initial beacon transmission
    - Angular rate estimation
    - Magnetic detumbling
    - Battery stabilization

4.2.3 Commissioning Mode

Objective

Verify subsystem health and calibrate spacecraft.

- ❖ Activities
    - ADCS calibration
    - Power system verification
    - Payload checkout
    - Communication testing
    - Orbit verification
    - Initial imaging tests
- ❖ Duration
    - Approximately 1–2 weeks

4.2.4 Science Mode

Objective

Acquire multispectral Earth imagery.

This table defines the primary operational mode of the spacecraft during precision agriculture and Earth observation missions. The payload operates in active imaging mode while the spacecraft maintains nadir-pointing orientation with attitude control accuracy better than 0.01 degrees to support high-quality multispectral image acquisition.

Table 4 3.Science Mode Characteristics Table

Parameter	Description
Payload	ACTIVE
Pointing	Nadir pointing
ADCS Accuracy	<0.01°
Data Storage	ACTIVE
Communication	Delayed until ground pass

- Activities
    - Scheduled agricultural imaging
    - NDVI-oriented multispectral acquisition
    - Long-term crop monitoring
    - Vegetation health assessment
    - Seasonal agricultural surveys
    - Secondary disaster observation missions
    - Image compression and onboard storage

4.2.5 Communication Mode

Objective:

Transmit telemetry and payload data to the ground station.

This table presents the spacecraft configuration during telemetry transmission and payload data downlink operations. During this mode, the spacecraft activates a high-gain communication link and reorients itself toward the ground station to maximize communication efficiency and link reliability. The operational design combines S-band downlink capability for high-rate payload data transfer with UHF uplink channels for command reception and mission updates. The activities defined in this mode, including telemetry transfer, payload data transmission, command upload, and timeline synchronization, demonstrate an understanding of CubeSat communication architectures and efficient management of spacecraft-ground interactions during limited ground station visibility windows.

Table 4 4.Communication Mode Characteristics Table

Parameter	Description

High Gain Link	ACTIVE
Payload	Usually, OFF
ADCS	Ground station pointing
Downlink	S-band
Uplink	UHF

- Activities
  - Telemetry downlink
  - Payload data transfer
  - Command upload
  - Mission timeline updates

4.2.6 Safe Mode

Objective

Protect spacecraft after anomaly detection.

This table defines the spacecraft’s fault-protection and anomaly recovery operational mode. In this emergency configuration, all nonessential systems such as the payload and onboard data handling processes are suspended to minimize power consumption and protect spacecraft integrity. The spacecraft autonomously performs Sun-pointing maneuvers to maximize solar power generation while maintaining only essential telemetry communication with the ground segment.

Table 4 5.Safe Mode Characteristics Table

Parameter	**Description**
Payload	OFF
ADCS	Sun pointing
Communication	Essential telemetry only
Power Usage	Minimized
Data Handling	Suspended

- Entry Triggers
  - Low battery voltage
  - ADCS fault

  - Thermal anomaly
  - Communication loss
  - Software fault
- ❖ Recovery Method
  - Ground-commanded recovery
  - Autonomous recovery sequence

4.3 Day in the Life of the Satellite

The spacecraft operates primarily as a precision agriculture monitoring platform. A nominal operational day is summarized below.

❖ Eclipse Phase

- Spacecraft operates on battery power
- Payload remains OFF to conserve energy
- Housekeeping telemetry collected

❖ Sunlit Orbit Segment

- Solar panels recharge batteries
- Spacecraft prepares for agricultural imaging
- ADCS transitions to nadir-pointing mode

❖ Agricultural Imaging Pass

- Payload activated over predefined agricultural regions
- Multispectral imagery acquired for vegetation analysis
- NDVI-compatible spectral data collected
- Data stored in onboard memory

❖ Onboard Processing Phase

- Image compression performed
- Priority agricultural datasets organized
- Telemetry packets prepared

❖ Ground Station Pass

- S-band downlink activated
- Agricultural data transmitted to ground station
- Updated imaging schedules uploaded
- Secondary disaster-response tasking uploaded if required

This operational cycle repeats several times per day depending on orbital geometry and target visibility.

4.4 Ground Segment Concept

The ground segment includes:

This table describes the ground segment architecture developed to support command, communication, and mission data utilization throughout spacecraft operations. The operational framework includes a mission control station for spacecraft command and monitoring, UHF and S-band ground antennas for communication, a dedicated data processing center for image analysis, and end-user interfaces serving agricultural and disaster-management authorities.

Table 4 6.Ground Segment Table

Element	Function
Mission Control Station	Command & control
Ground Antenna	UHF/S-band communications
Data Processing Center	Image processing
End Users	Agricultural/disaster authorities

The mission uses:

- UHF uplink for commands
- S-band downlink for payload data

D

5 Preliminary Sizing

Sizing estimation is based on:

- Historical CubeSat data [11][10][9][8]
- Commercial CubeSat subsystem references

5.1 Preliminary Mass Budget

This table presents the preliminary spacecraft mass allocation for all major subsystems within the 3U CubeSat architecture. The analysis demonstrates the ability to perform early-phase spacecraft sizing and subsystem allocation while maintaining compatibility with standard CubeSat launch constraints. The final estimated spacecraft mass of 4.30 kg confirms that the mission remains within the typical allowable launch mass range for a 3U CubeSat platform and therefore satisfies preliminary feasibility requirements.

Table 5 1.Preliminary Mass Budget

Subsystem	Estimated Mass (kg)
Structure & Mechanisms	0.60
Payload	1.10
EPS	0.45
Batteries	0.35
Solar Panels	0.30
ADCS	0.55
OBC	0.15
Communication System	0.35
Harness & Margins	0.45
Total Estimated Mass	4.30 kg

5.1.1 Mass Feasibility Assessment

Typical 3U CubeSat launch limits:

- 4–5 kg

Estimated spacecraft mass:

- 4.3 kg

Result:

- Positive mass margin exists
- Mission is feasible within launch constraints

5.2 Preliminary Power Budget

5.2.1 Power Architecture

The spacecraft uses:

- Deployable solar panels
- Lithium-ion battery pack
- Regulated EPS bus

Estimated average orbit power generation:

- 18–24 W (sunlit)

5.2.2 Operational Power Modes

This table defines the average spacecraft power consumption across different operational modes including safe mode, detumbling, science operations, communication mode, and nominal standby conditions. The analysis was developed to evaluate mission energy distribution and identify peak operational loads during mission execution. Science mode was identified as the most power-intensive operational phase due to simultaneous payload activity, attitude stabilization, and onboard data handling processes. The table demonstrates an understanding of spacecraft operational power profiling and mission energy management by linking subsystem activity levels to mode-dependent electrical consumption throughout the orbital cycle.

Table 5 2.Operational Power Modes

Mode	Average Power
Safe Mode	4 W
Detumbling	6 W
payload Mode	16 W
Communication Mode	14 W
Idle/Nominal	8 W

5.2.3 Subsystem Power Budget

This table provides a detailed subsystem-level power analysis by separating average and peak power consumption values for the payload, ADCS, onboard computer, communications system, EPS losses, and thermal subsystems. The budgeting process supports preliminary EPS sizing and enables identification of transient peak loads during simultaneous subsystem operation. The calculated total power demand establishes the baseline for evaluating solar array capability, battery sizing, and operational feasibility during all mission phases.

Table 5 3.Subsystem Power Budget

Subsystem	Average Power (W)	Peak Power (W)
Payload	6	9
ADCS	2.5	4
OBC[1]	1.5	2
Communications	3	8
EPS Losses	1	1.5
Thermal & Misc.	1	2
Total	15 W	26.5 W

5.2.4 Solar Array Capability

Assuming:

- Triple-junction GaAs solar cells
- 28–32% efficiency

[1] Onboard Computer

- Deployable panel configuration

Estimated available generation:

This table summarizes the estimated electrical power generation capability of the spacecraft solar arrays under average and peak sunlit conditions. The analysis assumes the use of triple-junction GaAs solar cells with deployable panel configurations and conversion efficiencies between 28% and 32%. The generated power estimates were compared directly with spacecraft operational consumption to evaluate mission feasibility and available power margin during orbital operations.

The power margin relationship used during the analysis is:

$$Power\ Margin = \frac{P_{generated} - P_{required}}{P_{required}} \times 100\%$$

This relationship was used to verify that the spacecraft generates sufficient electrical power during science operations while maintaining operational margin for subsystem uncertainties, battery charging, and temporary peak loads. The resulting positive power margin confirms that the proposed EPS architecture is operationally feasible for the mission profile.

Table 5 4.Solar array generated Power

Condition	Generated Power
Average Sunlit	20 W
Peak Sunlit	24 W

5.2.5 Power Feasibility Assessment

Science mode average consumption:

- 15 W

Average generation:

- 20 W

Power Margin:

- Approximately 30–35%

Result:

- Positive power margin
- Mission operationally feasible

5.3 Preliminary Link Budget

5.3.1 Communication Architecture

This table defines the preliminary communication architecture of the mission by assigning UHF frequencies for uplink commands and S-band frequencies for payload data downlink operations. The communication design reflects a standard CubeSat architecture optimized for low-rate command transmission and higher-rate multispectral imagery transfer. The selection of separate uplink and downlink frequency bands demonstrates an understanding of communication system optimization, bandwidth allocation, and operational reliability for Earth observation CubeSat missions.

Table 5 5.Preliminary Link Budget

Link	Frequency Band
Uplink	UHF
Downlink	S-band

5.3.2 Communication Assumptions

This table summarizes the key assumptions used during preliminary communication link analysis, including orbital altitude, ground station elevation angle, operating frequency, antenna gains, transmitter power, and data rate requirements. These assumptions establish the baseline operational environment for evaluating communication feasibility and expected downlink performance. The table demonstrates an understanding of the parameters influencing spacecraft communication performance and provides the analytical foundation required for preliminary link budget estimation and communication system sizing.

Table 5 6.Communication Assumptions Table

Parameter	Value
Orbit Altitude	650 km
Ground Station Elevation	20° minimum
Downlink Frequency	2.2 GHz
Data Rate	1–2 Mbps
Ground Antenna Gain	20 dBi
Spacecraft Antenna Gain	5 dBi

TX Power	2 W

5.3.3 Simplified Link Budget

This table presents the preliminary communication link budget analysis used to evaluate the feasibility of payload data transmission between the spacecraft and the ground station. Parameters including transmitter power, antenna gains, free-space path loss, atmospheric losses, and receiver sensitivity were incorporated to estimate received signal strength and communication margin.

The simplified communication relationship used during the analysis is:

$$P_r = P_t + G_t + G_r - L_p - L_a$$

where P_r represents received power, P_t is transmitter power, G_t and G_r are transmitter and receiver antenna gains, L_p is path loss, and L_a represents atmospheric and pointing losses. The resulting positive link margin of approximately 9 dB demonstrates that the selected communication architecture is capable of supporting reliable CubeSat-class payload downlink operations under the defined mission conditions.

Table 5 7.Simplified Link Budget

Parameter	Value
TX Power	+33 dBm
Spacecraft Antenna Gain	+5 dBi
Path Loss	-156 dB
Ground Antenna Gain	+20 dBi
Atmospheric/Pointing Losses	-3 dB
Received Power	-101 dBm
Receiver Sensitivity	-110 dBm
Link Margin	9 dB

5.3.4 Link Feasibility Assessment

The preliminary communication analysis shows:

- Positive communication margin
- Feasible payload downlink

- Adequate margin for CubeSat-class radios

The selected architecture is therefore acceptable for Phase A.

5.4 Preliminary ADCS Requirements

Because imaging quality strongly depends on pointing stability, the following preliminary ADCS requirements are defined.

This table defines the preliminary attitude determination and control system performance requirements necessary to support high-quality multispectral imaging operations. Parameters including pointing accuracy, pointing stability, detumbling duration, actuator configuration, and sensor selection were established based on payload imaging requirements and operational constraints. The requirement for sub-0.01-degree pointing accuracy reflects the sensitivity of Earth observation imaging missions to spacecraft attitude stability and image blur. The selected architecture using reaction wheels, magnetorquers, Sun sensors, magnetometers, and star trackers demonstrates an understanding of practical CubeSat ADCS design and subsystem integration for precision imaging missions.

Table 5 8.Preliminary ADCS Requirements

Requirement	Value
Pointing Accuracy	<0.01°
Pointing Stability	<0.001°/s
Detumbling Time	<3 orbits
Control Actuators	Reaction wheels + magnetorquers
Sensors	Sun sensors + magnetometer + star tracker

The selected requirements are achievable using existing CubeSat ADCS solutions.

5.5 Data Handling Concept

This table defines the onboard data management architecture developed for multispectral imagery storage, compression, prioritization, and transmission. The operational concept includes onboard memory capacity between 32 and 64 GB, CCSDS-compatible compression methods, and priority-based data management strategies for disaster-response imagery and spacecraft telemetry. The prioritization of emergency disaster imagery over routine agricultural observations demonstrates an understanding of mission flexibility and operational responsiveness during critical events.

Table 5 9.Data Handling Concept

Parameter	Value
Onboard Storage	32–64 GB
Compression	CCSDS compatible
Priority Handling	Supported
Disaster Data Priority	HIGH

The spacecraft prioritizes:

8. Emergency disaster imagery
9. Health telemetry
10. Agricultural survey imagery

5.6 Risk & Feasibility Assessment

5.6.1 Major Technical Risks

This table presents the preliminary technical risk assessment for the mission by identifying critical spacecraft risks, evaluating their operational impacts, and defining corresponding mitigation strategies. Risks including power deficit, ADCS instability, data overflow, thermal variation, and communication outage were analyzed together with mitigation methods such as deployable solar panels, reaction wheel stabilization, onboard compression techniques, passive thermal control, and multiple communication passes.

Table 5 10.Major Technical Risks

Risk	Impact	Mitigation
Power deficit	Mission interruption	Deployable panels + duty cycling
ADCS instability	Blurred imagery	Reactionwheel stabilization
Data overflow	Data loss	Compression+ prioritization

Thermal variation	Payload degradation	Passive thermal design
Communication outage	Delayed delivery	Multiple daily passes

5.7 Feasibility Summary

This table summarizes the overall technical feasibility assessment of the proposed 3U CubeSat mission across major engineering domains including payload integration, orbital configuration, power budget, mass budget, communication performance, ADCS capability, and CubeSat compatibility. Each subsystem area was evaluated against preliminary mission requirements and operational constraints to determine whether acceptable performance margins existed. The final feasibility assessment demonstrates that the spacecraft architecture satisfies Phase A mission requirements and provides a technically achievable baseline for subsequent preliminary design and subsystem refinement activities.

Table 5 11.Feasibility Summary

Area	Assessment
Payload Feasibility	FEASIBLE
Orbit Feasibility	FEASIBLE
Power Budget	POSITIVE MARGIN
Mass Budget	POSITIVE MARGIN
Link Budget	POSITIVE MARGIN
ADCS Capability	FEASIBLE
CubeSat Compatibility	FEASIBLE

5.8 Preliminary Requirements Review (PRR) Conclusion

The Phase A study demonstrates that the proposed 3U CubeSat mission is technically feasible within CubeSat-class constraints.

The mission has been successfully defined as a precision agriculture Earth observation mission with secondary disaster monitoring capability.

Key outcomes include:

- A feasible multispectral payload optimized for vegetation monitoring
- A mission architecture centered on precision agriculture applications

- Operationally valid ConOps architecture
- Positive preliminary power margin
- Positive preliminary communication link margin
- Acceptable mass budget within 3U constraints
- Achievable ADCS performance for agricultural imaging missions

The selected Sun-Synchronous Orbit, multispectral payload, and operational strategy provide appropriate revisit capability (mean 7 days) and illumination consistency for agricultural monitoring.

Secondary disaster-response functionality can also be supported through flexible mission retasking without significantly affecting the primary mission objectives.

The mission concept therefore successfully satisfies Preliminary Requirements Review (PRR) level feasibility requirements and is ready to proceed toward:

- Phase B preliminary design
- Detailed subsystem trade studies
- Payload optimization
- Thermal and structural analyses
- Detailed ADCS simulations
- Detailed communications architecture refinement

6 Conclusion

The Phase A study successfully established a preliminary system architecture for a 3U precision-agriculture CubeSat mission operating in a 650 km Sun-Synchronous Orbit. The mission concept was developed around a multispectral Earth-observation payload optimized for NDVI generation, vegetation-health assessment, crop monitoring, and seasonal agricultural analysis. The selected orbital configuration, operational concept, and subsystem architecture collectively support periodic revisits and stable illumination conditions required for agricultural remote-sensing applications. In addition to the primary agricultural mission, the spacecraft architecture also supports secondary disaster-monitoring objectives including flood observation, drought monitoring, and wildfire assessment through flexible mission retasking and onboard data prioritization.

Preliminary subsystem sizing demonstrated that the mission remains broadly feasible within CubeSat-class physical and operational limitations. The total estimated spacecraft mass of approximately 4.3 kg remains within standard 3U launch constraints, while the preliminary electrical power analysis indicates an average sunlit generation capability of approximately 20 W against a science-mode consumption of nearly 15–16 W, resulting in a positive operational power margin. Similarly, the communication architecture using UHF uplink and S-band downlink frequencies demonstrated a preliminary link margin of approximately 9 dB, supporting the feasibility of multispectral image transmission to the ground segment. The mission operational concept, including launch, detumbling, commissioning, science, communication, and safe operational modes, also demonstrated a coherent mode-based spacecraft management architecture suitable for autonomous CubeSat operations.

Although the overall mission concept was shown to be technically promising, several engineering challenges were identified that require further refinement in future design phases. The primary technical concern involves the precision attitude determination and control requirements necessary for stable multispectral imaging, particularly with respect to pointing accuracy, jitter reduction, and long-duration nadir-pointing performance. In addition, payload packaging within a 3U spacecraft volume remains a significant design constraint due to the simultaneous integration of imaging optics, communication systems, power subsystems, and attitude-control hardware. Future development phases should therefore focus on detailed ADCS simulations, thermal and structural analyses, payload integration optimization, vibration and jitter assessment, and refined subsystem trade studies to validate long-term mission feasibility and operational performance.

References

[1] https://simera-sense.com/news/multispectral-cubesat-imagers-understanding-the-benefits-and-limitations/#::text=SPATIAL%20RESOLUTION

[2] https://www.mdpi.com/1424-8220/24/9/2885

[3] https://dragonflyaerospace.com/products/#::text=12U%20form%20factor,SSA%29%20applications

[4] https://technology.nasa.gov/patent/GSC-TOPS-138#::text=This%20dual%20band%20infrared%20imaging,pitch%2C%20640%20x%20512%20pixel

[5] https://www.cubesat.org/descriptions#::text=reliability%20small%20satellite%20and%20cubesat,SAR%20radar%20for%20small%20satellites

[6] https://www.sciencedirect.com/science/article/abs/pii/S0094576521003672#::text=

[7] https://satcatalog.s3.amazonaws.com/components/768/SatCatalog_-_Simera_Sense_-_MultiScape100_CIS_-_Datasheet.pdf?lastmod=20210714222315#::text=The%20MultiScape100%20CIS%20is%20a,infrared%20(VNIR)%20spectral%20range.

[8] https://arxiv.org/pdf/1706.06967

[9] https://asteroid.arizona.edu/optical_nav_SRS_jth_public.pdf

[10] https://jatm.com.br/jatm/article/view/281

[11] https://www.researchgate.net/publication/355601167_Topology_optimization_and_additive_manufacturing_applied_to_a_camera_bracket_for_a_3U_CubeSat